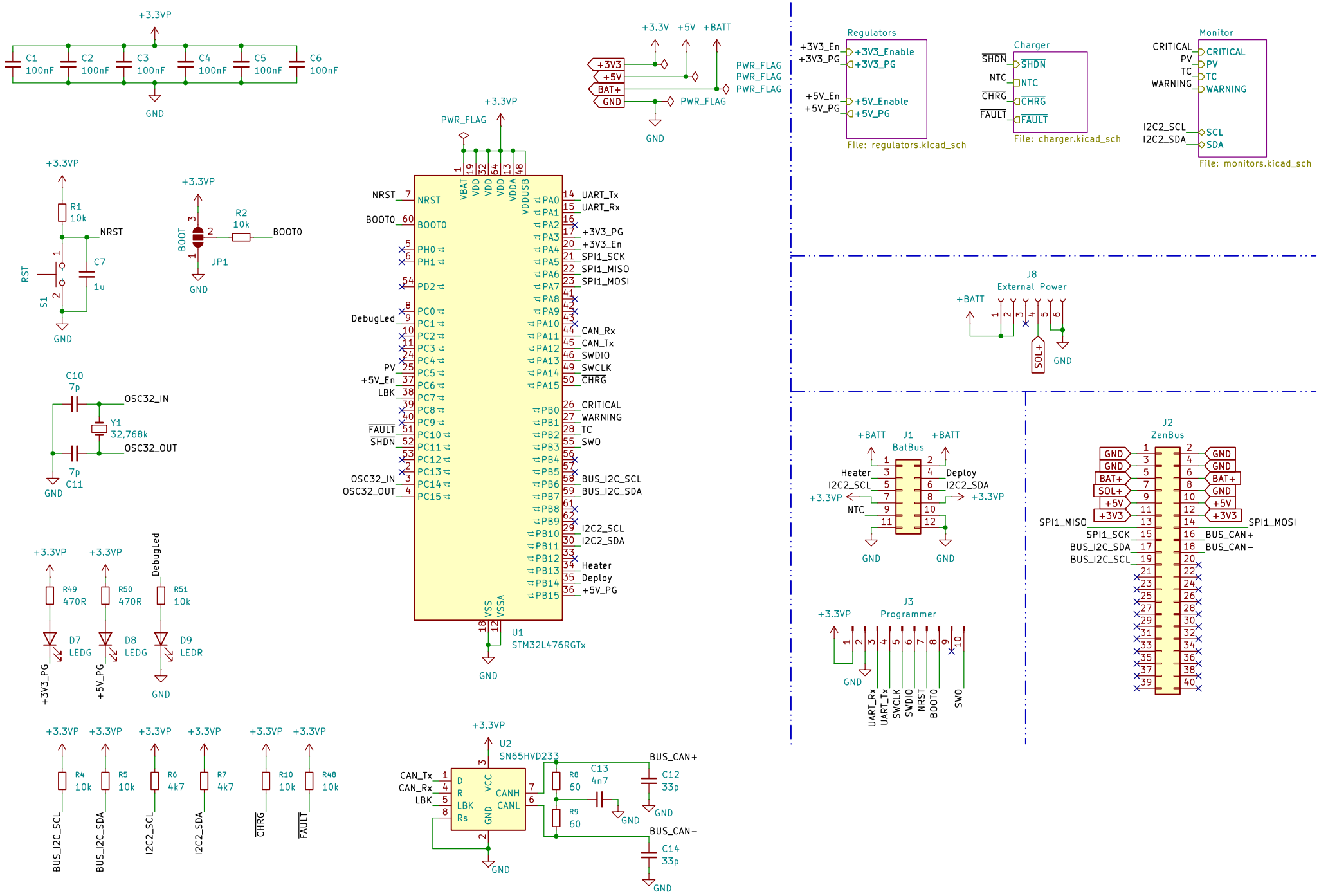
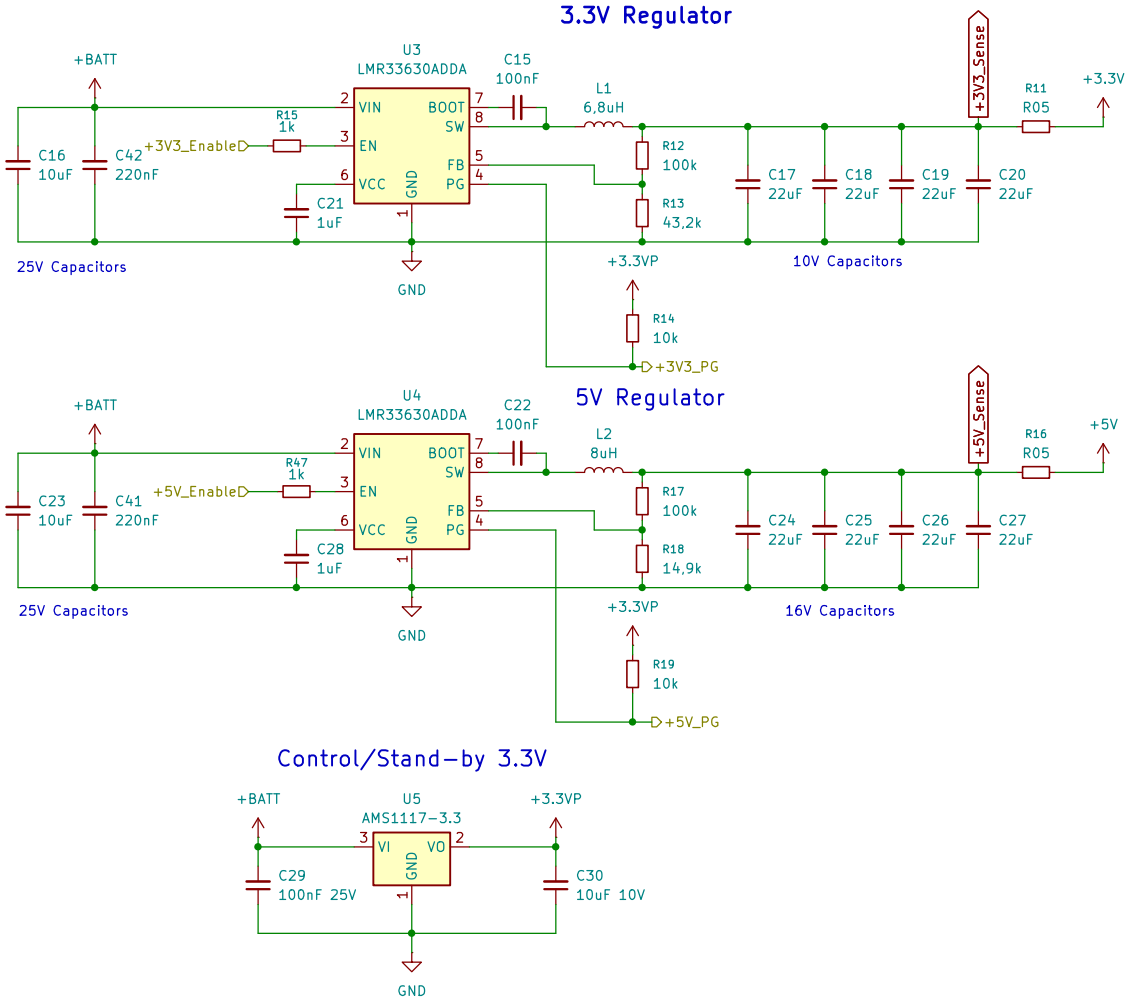
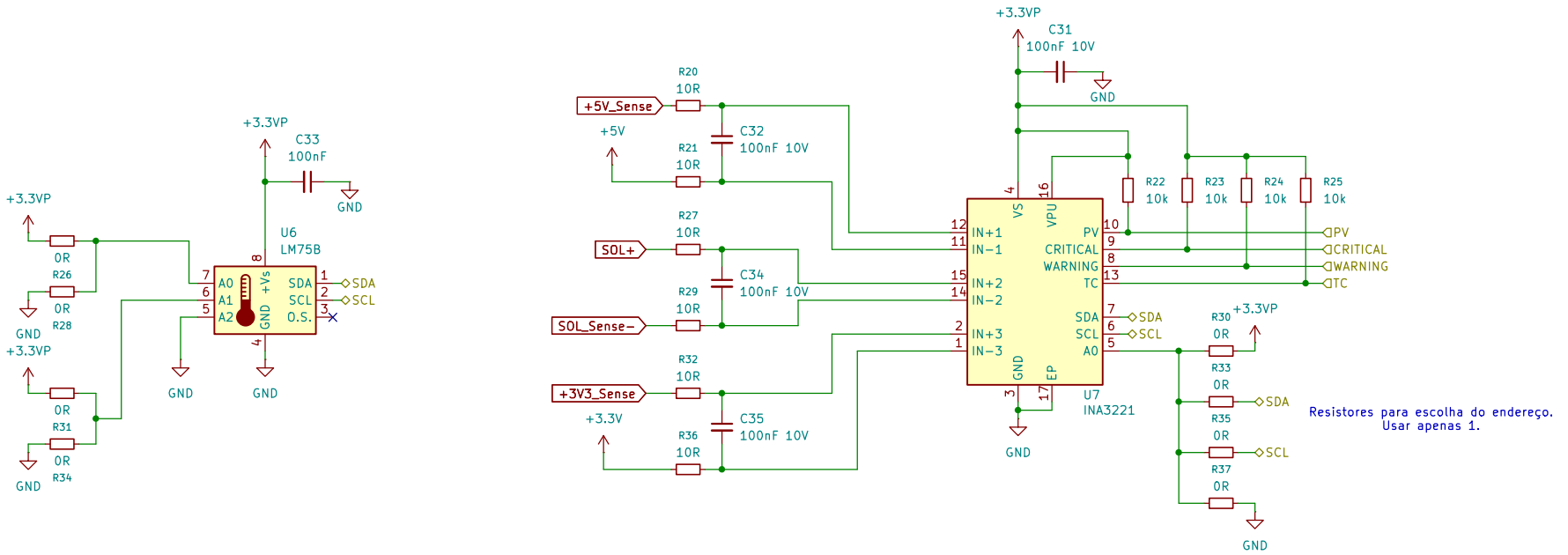
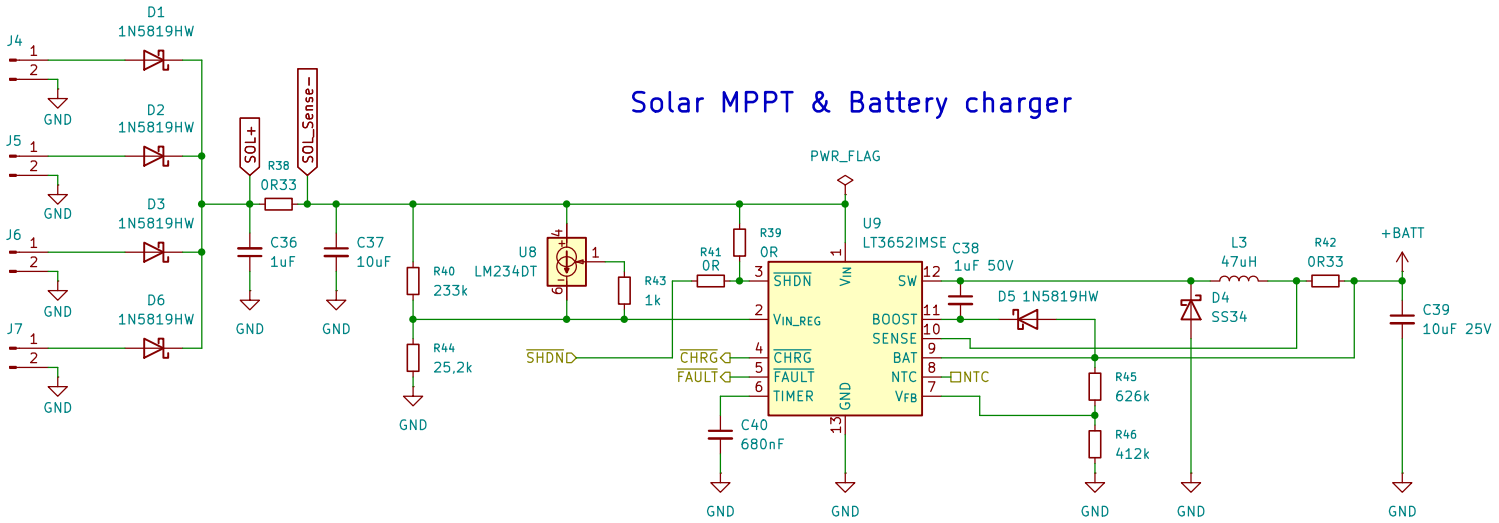

+3.3VP
C1 100nF
C2 100nF
C3 100nF
C4 100nF
C5 100nF
C6 100nF
GND
+3.3V +5V +BATT
+3V3
+5V
BAT+
GND
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
Regulators
+3V3_En
+3V3_PG
+3V3_Enable
+3V3_PG
+5V_En
+5V_PG
+5V_Enable
+5V_PG
File: regulators.kicad_sch
Charger
SHDN
NTC
CHRG
FAULT
File: charger.kicad_sch
Monitor
CRITICAL
PV
TC
WARNING
I2C2_SCL
I2C2_SDA
SCL
SDA
File: monitors.kicad_sch
R1 10k
NRST
C7 1u
RST
S1
GND
R2 10k
BOOT0
JP1
BOOT
GND
U1
STM32L476RGTx
NRST
BOOT0
PH0
PH1
PD2
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
VBAT
VDD
VDD
VDD
VDDA
VDDUSB
VSS
VSSA
UART_Tx
UART_Rx
+3V3_PG
+3V3_En
SPI1_SCK
SPI1_MISO
SPI1_MOSI
CAN_Rx
CAN_Tx
SWDIO
SWCLK
CHRG
CRITICAL
WARNING
TC
SWO
BUS_I2C_SCL
BUS_I2C_SDA
I2C2_SCL
I2C2_SDA
Heater
Deploy
+5V_PG
DebugLed
PV
+5V_En
LBK
FAULT
SHDN
OSC32_IN
OSC32_OUT
C10 7p
Y1 32,768k
C11 7p
OSC32_IN
OSC32_OUT
GND
J8
External Power
+BATT
SOL+
GND
J1
BatBus
+BATT
+BATT
Heater
I2C2_SCL
+3.3VP
NTC
Deploy
I2C2_SDA
+3.3VP
GND
GND
J2
ZenBus
GND
GND
BAT+
SOL+
+5V
+3V3
SPI1_MISO
SPI1_MOSI
SPI1_SCK
BUS_I2C_SDA
BUS_I2C_SCL
BUS_CAN+
BUS_CAN−
J3
Programmer
+3.3VP
GND
UART_Rx
UART_Tx
SWCLK
SWDIO
NRST
BOOT0
SWO
R49 470R
R50 470R
R51 10k
D7 LEDG
D8 LEDG
D9 LEDR
+3V3_PG
+5V_PG
GND
DebugLed
R4 10k
R5 10k
R6 4k7
R7 4k7
R10 10k
R48 10k
BUS_I2C_SCL
BUS_I2C_SDA
I2C2_SCL
I2C2_SDA
CHRG
FAULT
U2
SN65HVD233
CAN_Tx
CAN_Rx
LBK
D
R
LBK
Rs
VCC
GND
CANH
CANL
R8 60
R9 60
C13 4n7
C12 33p
C14 33p
BUS_CAN+
BUS_CAN−
GND


3.3V Regulator
+BATT
U3
LMR33630ADDA
C15
100nF
L1
6,8uH
+3V3_Sense
R11
R05
+3.3V
R15
1k
+3V3_Enable
C16
10uF
C42
220nF
VIN
BOOT
SW
EN
FB
VCC
GND
PG
R12
100k
R13
43,2k
C17
22uF
C18
22uF
C19
22uF
C20
22uF
C21
1uF
25V Capacitors
GND
+3.3VP
10V Capacitors
R14
10k
+3V3_PG
5V Regulator
+BATT
U4
LMR33630ADDA
C22
100nF
L2
8uH
+5V_Sense
R16
R05
+5V
R47
1k
+5V_Enable
C23
10uF
C41
220nF
R17
100k
R18
14,9k
C24
22uF
C25
22uF
C26
22uF
C27
22uF
C28
1uF
25V Capacitors
GND
+3.3VP
16V Capacitors
R19
10k
+5V_PG
Control/Stand-by 3.3V
+BATT
U5
AMS1117-3.3
+3.3VP
VI
VO
GND
C29
100nF 25V
C30
10uF 10V
GND

I/V Monitor


+3.3VP
C31
100nF 10V
GND
R20
10R
+5V_Sense
+5V
C32
100nF 10V
R21
10R
R27
10R
SOL+
C34
100nF 10V
R29
10R
SOL_Sense−
R32
10R
+3V3_Sense
+3.3V
C35
100nF 10V
R36
10R
R22
10k
R23
10k
R24
10k
R25
10k
IN+1
IN−1
IN+2
IN−2
IN+3
IN−3
VS
VPU
GND
EP
PV
CRITICAL
WARNING
TC
SDA
SCL
A0
U7
INA3221
PV
CRITICAL
WARNING
TC
SDA
SCL
R30
0R
+3.3VP
R33
0R
SDA
R35
0R
SCL
R37
0R
GND
Resistores para escolha do endereço.
Usar apenas 1.
+3.3VP
C33
100nF
GND
U6
LM75B
A0
A1
A2
+Vs
GND
SDA
SCL
O.S.
SDA
SCL
+3.3VP
0R
R26
0R
R28
GND
+3.3VP
0R
R31
0R
R34
GND
GND
GND


Solar MPPT & Battery charger
D1
1N5819HW
D2
1N5819HW
D3
1N5819HW
D6
1N5819HW
J4
J5
J6
J7
GND
SOL+
SOL_Sense-
R38
0R33
C36
1uF
C37
10uF
R40
233k
R44
25,2k
U8
LM234DT
R43
1k
R41
0R
R39
0R
PWR_FLAG
U9
LT3652IMSE
SHDN
VIN_REG
CHRG
FAULT
TIMER
VIN
SW
BOOST
SENSE
BAT
NTC
VFB
C38
1uF 50V
D5 1N5819HW
L3
47uH
D4
SS34
R42
0R33
+BATT
C39
10uF 25V
R45
626k
R46
412k
C40
680nF
SHDND
CHRG
FAULT
NTC